

Data Science Capstone Project Report

SpaceX Falcon 9 First-Stage Landing Prediction

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GitHub repository (all completed notebooks and Python files):

<https://github.com/razerblade09/spacex-falcon9-capstone>

This single PDF contains both required deliverables:

- Completed presentation, all 19 slides — pages 2–20
- Written project report — pages 21–32

Required sections, by page

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The written report on pages 21–32 repeats every section above in long form, with the same figures and the GitHub link for each stage.

SpaceX Falcon 9 First-Stage Landing Prediction

Data Science Capstone Project Report

GitHub repository (all notebooks and Python files): <https://github.com/razerblade09/spacex-falcon9-capstone>

Executive Summary

Methods used: Data was collected via the SpaceX REST API and Wikipedia web scraping, cleaned and labeled, explored with visualization and SQL, mapped geospatially with Folium, and modeled with four classifiers (Logistic Regression, SVM, Decision Tree, KNN) tuned via GridSearchCV.

Key results: Across 90 Falcon 9 launches, the overall first-stage landing success rate was 67%. The best-performing model, Logistic Regression, reached 83% test accuracy in predicting landing outcome.

90

Launches analyzed

67%

Overall landing success rate

83.3%

Best model test accuracy

4

ML models compared

Conclusion: Landing success is highly predictable from launch parameters, meaning a competing launch provider could reliably estimate SpaceX's true reusable-launch cost advantage.

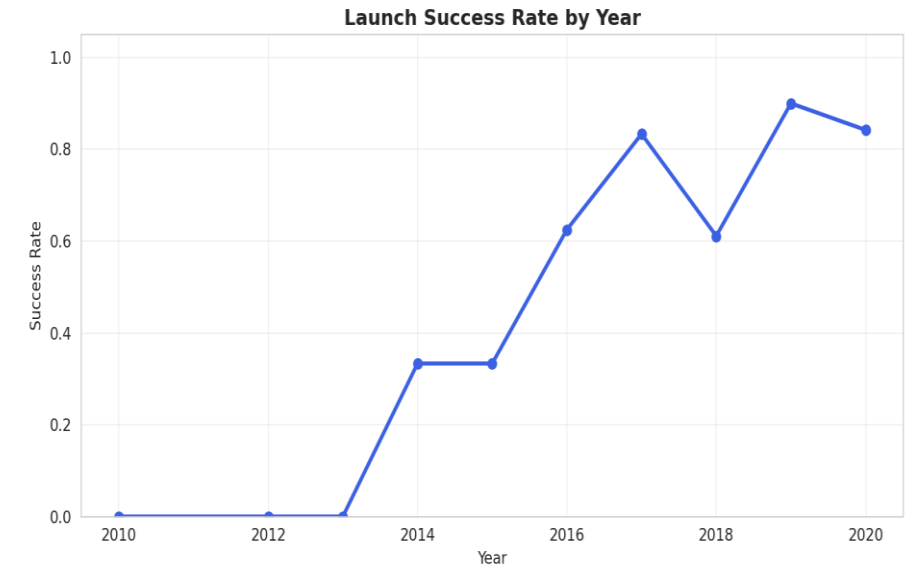
Introduction

SpaceX advertises Falcon 9 launches at \$62 million, versus \$165M+ from other providers — largely because SpaceX reuses the rocket's first stage instead of discarding it.

Problem statement: If the first stage does not land successfully, that launch's cost estimate changes substantially. This project builds a model to predict whether the first stage will land, using historical launch data — enabling accurate cost estimation for a competing bid.

Objectives:

- Collect and clean historical Falcon 9 launch data
- Identify factors correlated with landing success
- Visualize patterns across sites, orbits, and time
- Build and evaluate a landing-outcome classifier



Falcon 9 landing success climbed steadily as SpaceX matured its recovery process.

Data Collection – SpaceX API

Pipeline:

1

GET /v4/launches/past → all historical launches

2

Filter to single-core, single-payload Falcon 9 flights

3

For each launch, call /rockets, /launchpads, /payloads, /cores endpoints

4

Assemble into one row per launch: booster, orbit, site, outcome, mass

```
api.spacexdata.com/v4/launches/past
```

Result: 90 clean Falcon 9 launch records (2010–Nov 2020), each with booster version, payload mass, orbit, launch site, flight count, grid fins, reused flag, legs, landing pad, and outcome.

Falcon 1 launches were excluded — this project focuses only on the Falcon 9 landing-prediction problem.

Data Collection – Web Scraping

Pipeline:

- 1 Request the Wikipedia "List of Falcon 9 and Falcon Heavy launches" page
- 2 Parse HTML tables with BeautifulSoup, locate the wikitable elements
- 3 Extract column headers, then walk each row of launch data
- 4 Clean text (dates, booster serials, payload mass) into a dataframe

```
BeautifulSoup(response.text, 'html.parser')
```

This gives an independent, cross-checked source for launch records — used to validate the API dataset and fill in fields (like precise payload customer) the API doesn't expose.

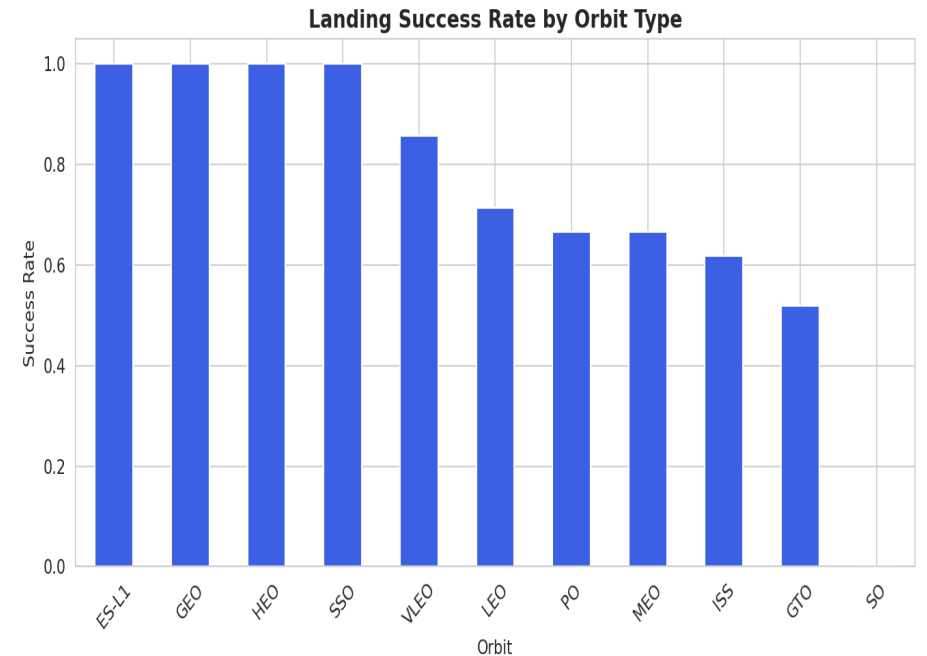
Key techniques: regex/Unicode cleanup for payload mass strings, and helper functions to pull booster version and landing status out of merged table cells.

Data Wrangling Methodology

Missing values: PayloadMass (5.6% missing) filled with column mean; LandingPad (28.9% missing) kept as-is since a null pad legitimately means "no ground/drone-ship landing occurred."

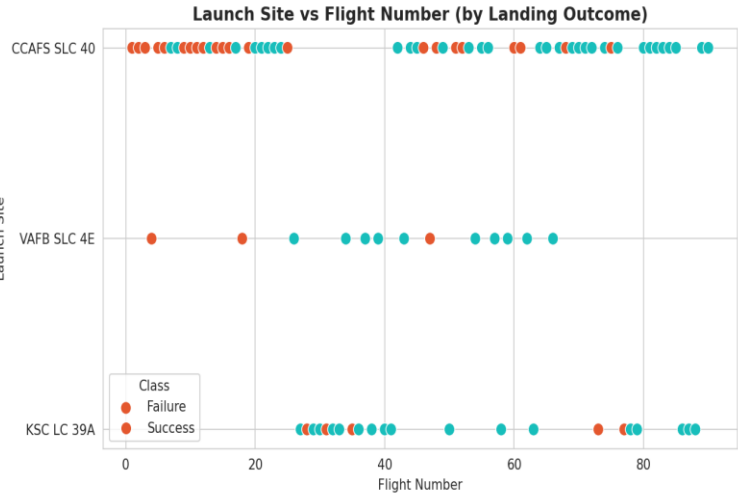
Label engineering: The raw Outcome field (e.g. "True ASDS", "False Ocean", "None None") was converted into a binary Class label — 1 if the booster landed successfully by any method, 0 otherwise.

Result: 90 launches → 60 successful landings (66.7%), 30 unsuccessful (33.3%)

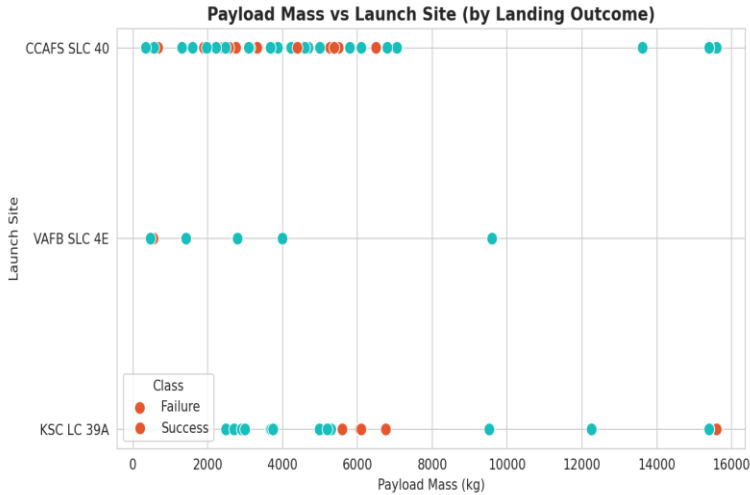


EDA with Data Visualization

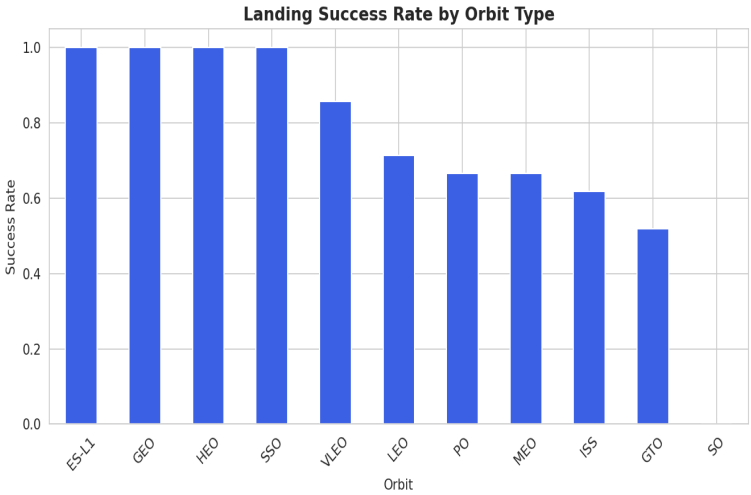
We plotted flight number, payload mass, orbit type, and launch year against landing outcome to surface which factors correlate with success.



Flight # vs Launch Site



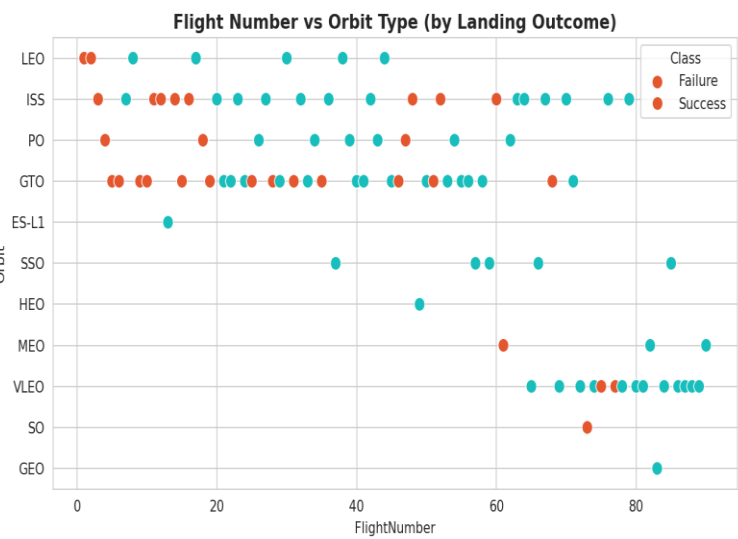
Payload vs Launch Site



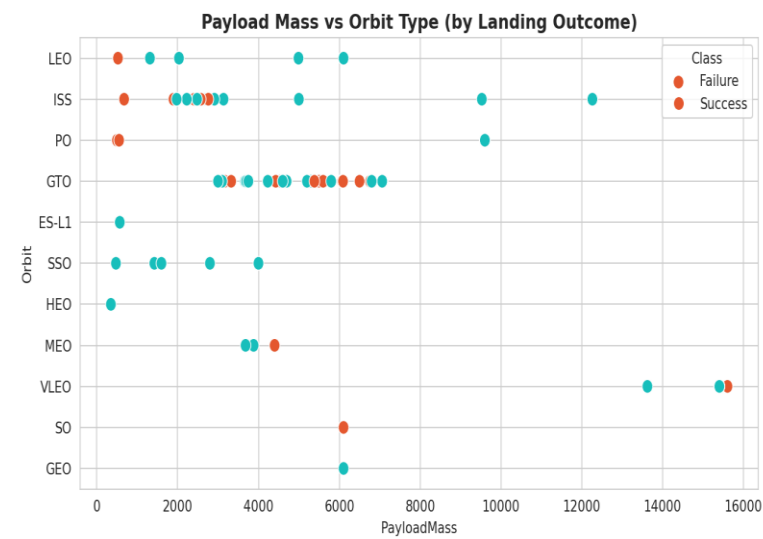
Success Rate by Orbit

Chart types used: scatter plots (flight #/payload vs site or orbit), bar charts (success rate by orbit), and a line chart (yearly trend, see Introduction slide).

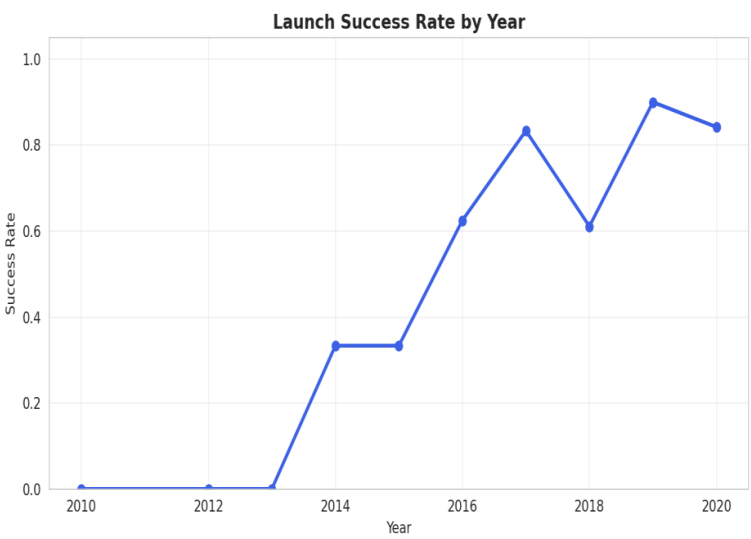
EDA Visualization — Orbit & Trend Detail



Flight # vs Orbit Type



Payload vs Orbit Type



Success Rate by Year

Takeaway: GTO and ISS-orbit missions (heavier, higher-energy) show lower success rates; success climbed year-over-year as SpaceX refined the recovery process, from ~0% in 2010–2013 to consistently near 100% by 2019–2020.

EDA with SQL

The scraped launch table was loaded into a SQL database (SQLite / IBM Db2) as SPACEXTABLE, then queried directly to answer specific business questions.

- ✓ Distinct launch sites used
- ✓ Total payload mass carried for NASA (CRS)
- ✓ Average payload mass for F9 v1.1 boosters
- ✓ Date of the first successful ground-pad landing
- ✓ Boosters that succeeded on a drone ship with 4,000–6,000 kg payload
- ✓ Count of each mission outcome (success/failure)
- ✓ Booster(s) carrying the maximum payload mass
- ✓ Failed drone-ship landings in 2015
- ✓ Ranking of landing outcomes between 2010–2017

EDA SQL Results — Sites & Payload

Launch Sites (4 distinct)

CCAFS LC-40 · CCAFS SLC-40 · KSC LC-39A · VAFB SLC-4E

Total payload for NASA (CRS): 45,596 kg

Avg. payload, F9 v1.1 boosters: 2,534.7 kg

Max payload carried: 15,600 kg (F9 B5)

Milestones

First successful ground-pad landing: 1 May 2017

2015 drone-ship landing failures: 2 (F9 v1.1 B1012, B1015 — CCAFS LC-40)

4,000–6,000 kg drone-ship successes: F9 FT B1021.2, B1022, B1026, B1031.2

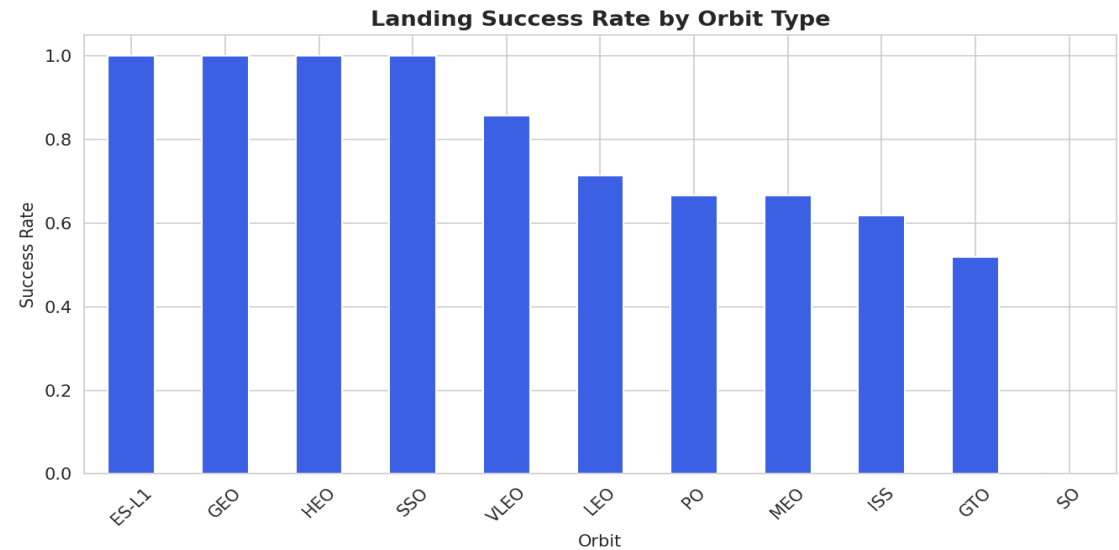
These figures ground the EDA charts in exact, query-verified numbers rather than visual estimates.

EDA SQL Results — Success Rates & Rankings

Landing outcome ranking, 2010–2017 (by count):

Landing Outcome	Count
Success	20
No attempt	10
Success (drone ship)	8
Success (ground pad)	6
Failure (drone ship)	4
Failure	3
Controlled (ocean)	3
Failure (parachute)	2

Mission outcome (2010–2020): 98 of 101 recorded outcomes were successful missions — first-stage recovery success is the harder, separate problem this model targets.

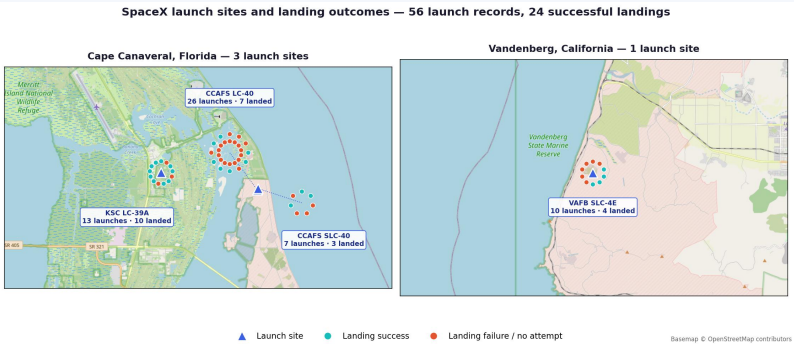


Interactive Visual Analytics

Two interactive tools let stakeholders explore the data themselves rather than reading static charts:

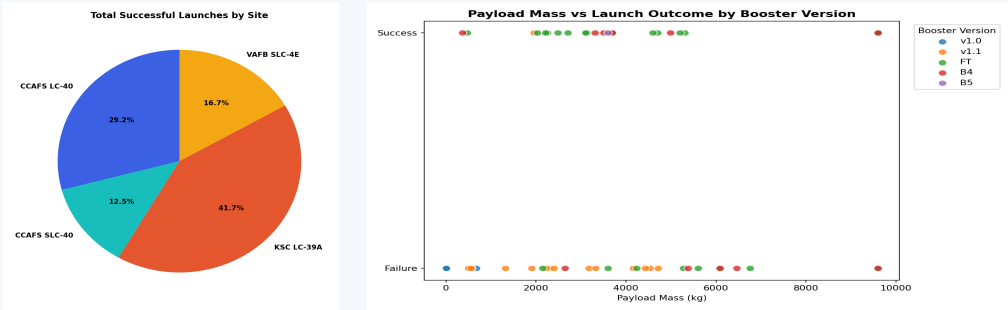
Folium Map

Launch-site markers, per-launch outcome markers (color-coded), and proximity distance lines (coastline, city, railway, highway).



Plotly Dash App

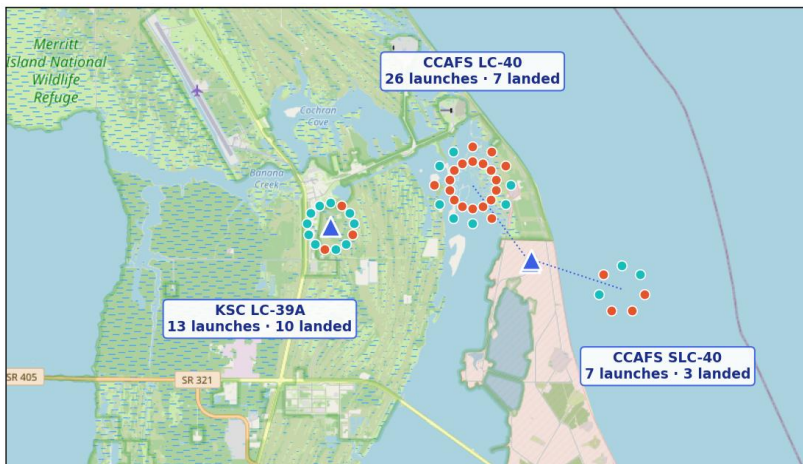
Dropdown to filter by launch site; live-updating success-rate pie chart and payload-vs-outcome scatter plot.



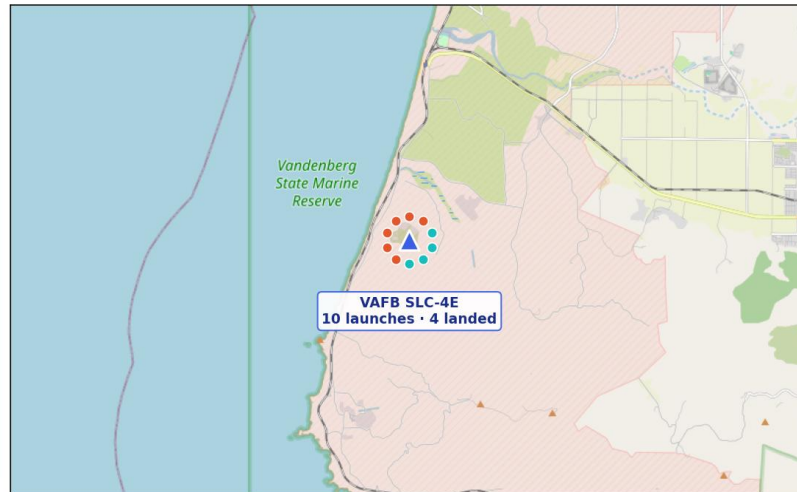
Launch Site Markers & Launch Records

SpaceX launch sites and landing outcomes — 56 launch records, 24 successful landings

Cape Canaveral, Florida — 3 launch sites



Vandenberg, California — 1 launch site



▲ Launch site ● Landing success ● Landing failure / no attempt

Basemap © OpenStreetMap contributors

All 4 launch sites plotted with markers.

Every individual launch (56 records) plotted as a marker cluster, colored green for a successful landing and red for a failed/no-attempt landing — letting you see which sites cluster more successes.

Observation: KSC LC-39A shows the densest green cluster — 10 landings from 13 launches (77%). CCAFS LC-40, the earliest and highest-volume pad, landed only 7 of 26 (27%): most of its records predate a working recovery process.

Proximity Analysis

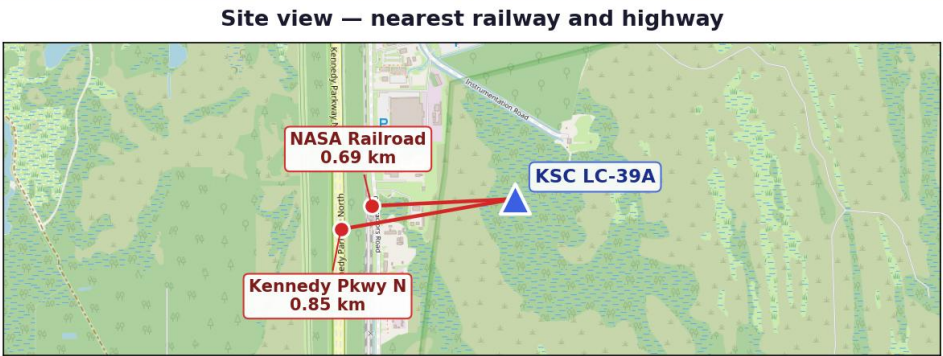
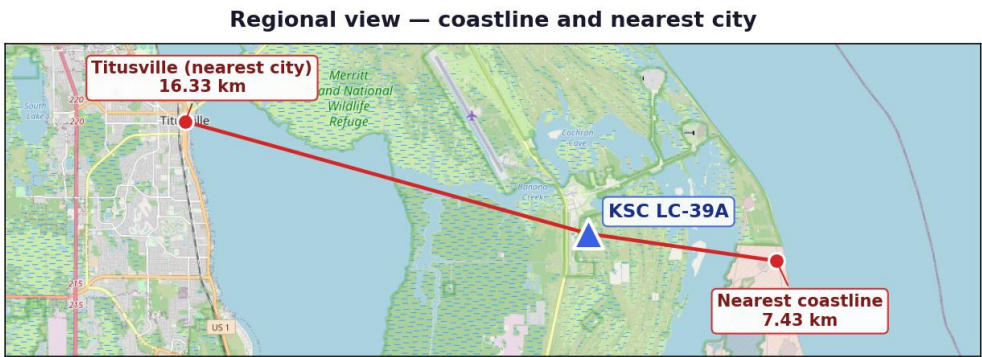
Using the haversine formula, we measured distance from KSC LC-39A to nearby geographic features:

Feature	Distance from Launch Site
Nearest coastline point	7.43 km
Nearest city (Titusville)	16.33 km
Nearest railway (NASA Railroad)	0.69 km
Nearest highway (Kennedy Pkwy N)	0.85 km

Why this matters: Launch sites are deliberately sited close to coastlines (for downrange safety over open water) and to rail and road links (boosters and stages arrive overland), but a safe distance from population centers — real siting constraints any competing launch provider would need to replicate.

City, railway and highway coordinates sourced from OpenStreetMap.

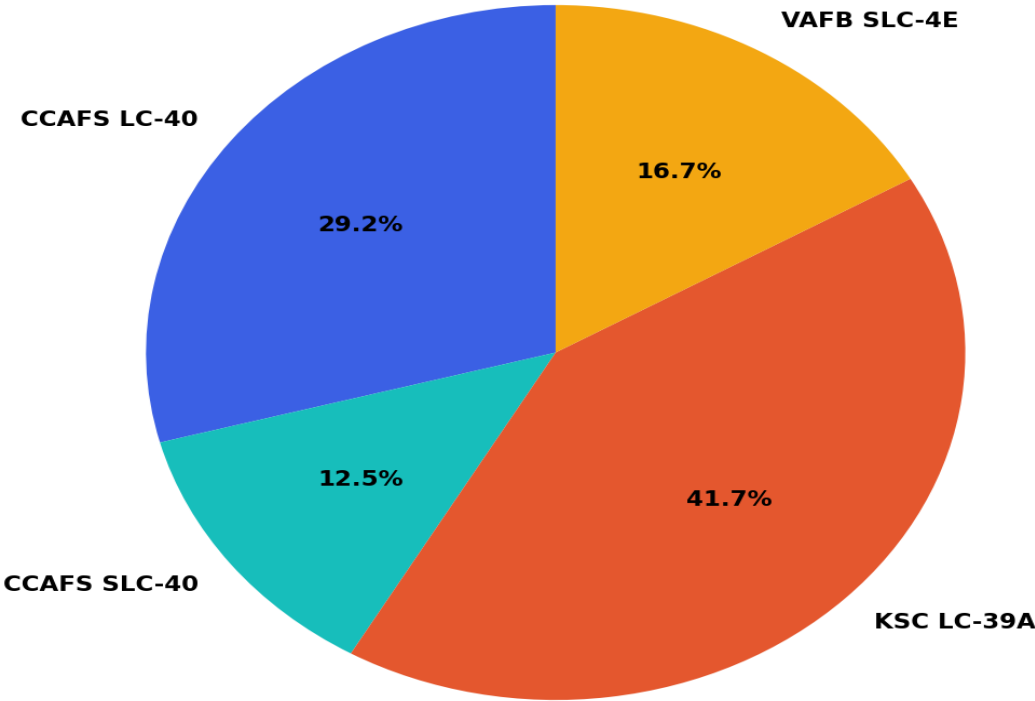
Proximity analysis — distances from KSC LC-39A



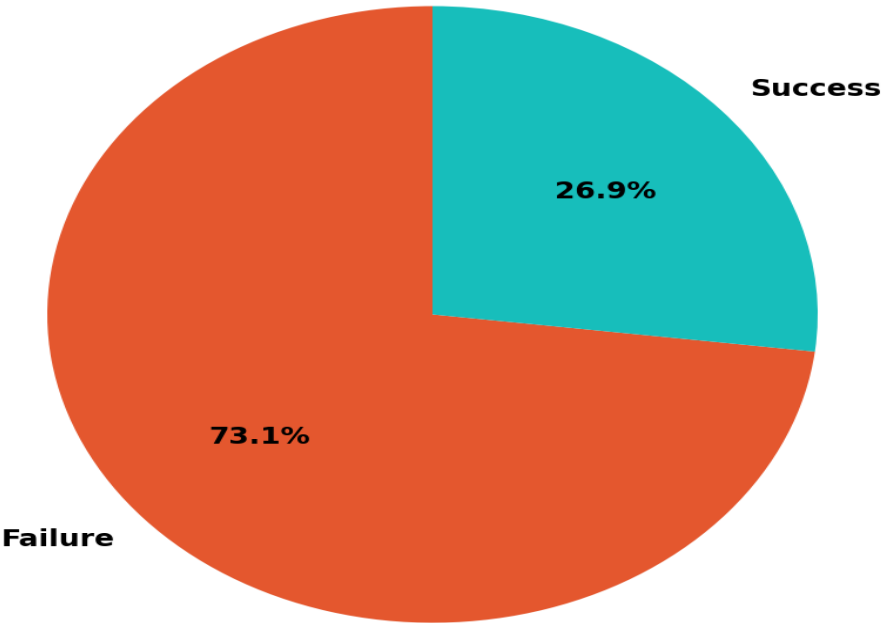
Basemap © OpenStreetMap contributors

Success Rate — Pie Charts

Total Successful Launches by Site

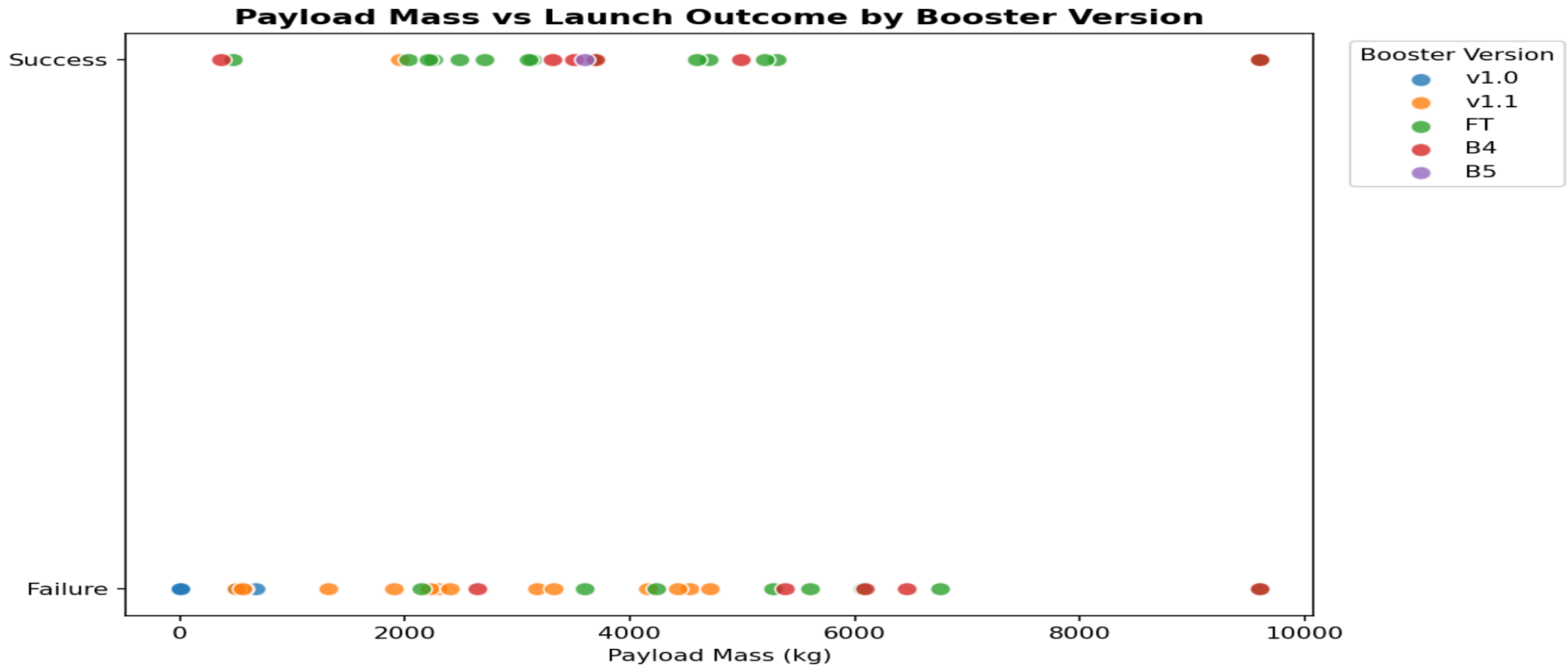


Success vs Failure - CCAFS LC-40



Left: share of total successful landings contributed by each site. Right: success-vs-failure split at the top-volume site (CCAFS LC-40) — filterable live via the dashboard dropdown.

Payload Mass vs. Launch Outcome



A RangeSlider (0–10,000 kg) lets users narrow the payload window; the scatter plot recolors by booster version category (v1.0 / v1.1 / FT / B4 / B5), revealing that newer booster generations succeed reliably across a much wider payload range than early versions.

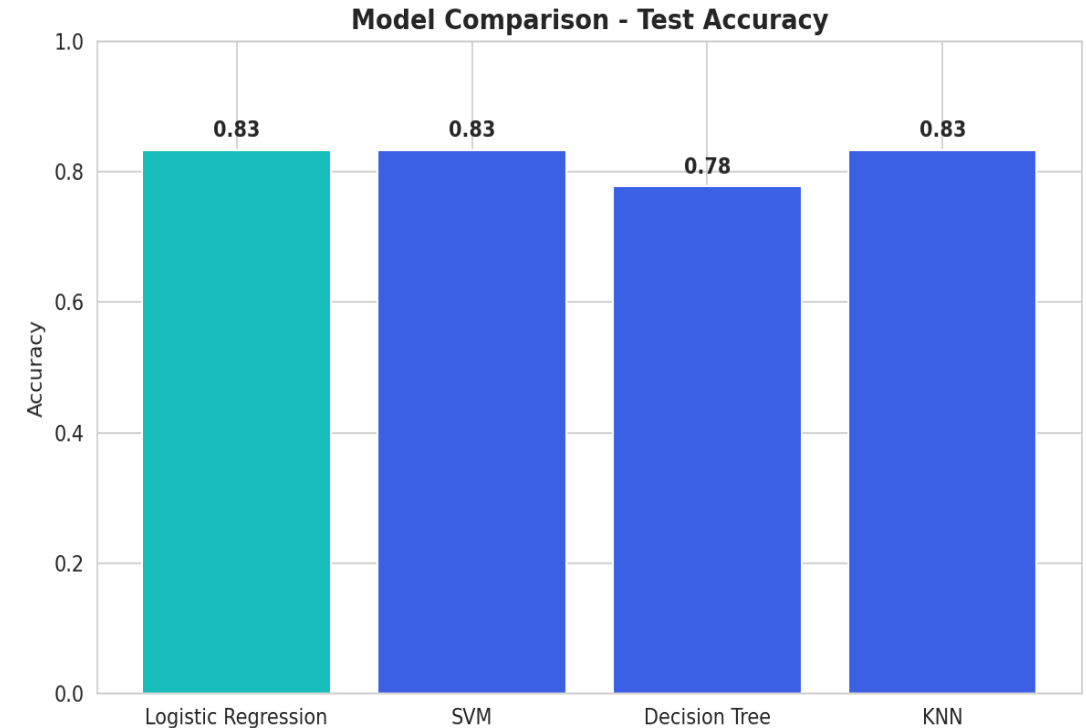
Model Development

Features: 80 one-hot-encoded columns (orbit, launch site, landing pad, booster serial) plus flight number, payload mass, flights, grid fins, reused, legs, and block — standardized with StandardScaler.

Split: 80% train / 20% test (72 / 18 launches), stratified via random_state=2.

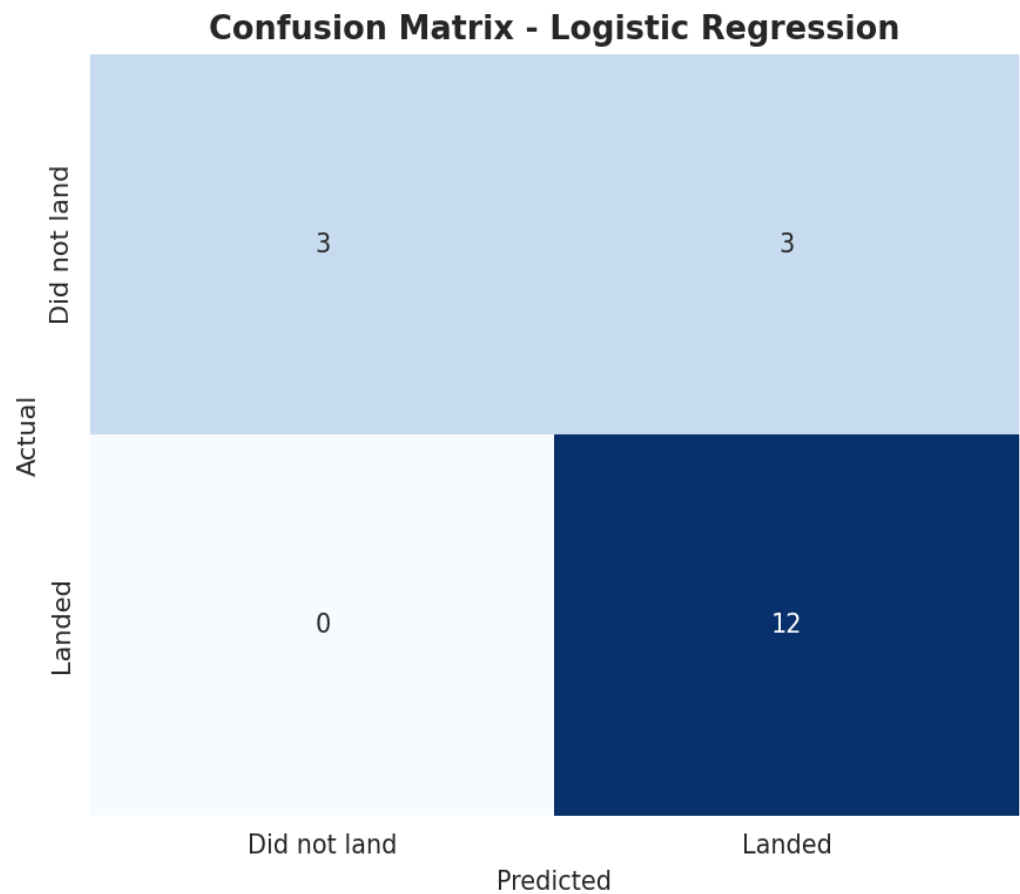
Models compared (each tuned via 10-fold GridSearchCV):

- Logistic Regression — C, penalty, solver
- Support Vector Machine — kernel, C, gamma
- Decision Tree — criterion, max_depth, min_samples_split
- K-Nearest Neighbors — n_neighbors, distance metric (p)



All four models converged near 83% test accuracy, with Decision Tree slightly behind — signal that the engineered features are strong and model choice matters less than feature quality here.

Evaluation Results & Confusion Matrix



Model	CV Score	Test Accuracy
Logistic Regression	0.821	0.833
SVM	0.834	0.833
KNN	0.834	0.833
Decision Tree	0.804	0.778

Reading the confusion matrix (18 test launches): 12 successful landings correctly predicted, 3 unsuccessful landings correctly predicted, 3 unsuccessful landings misclassified as successful, 0 successful landings missed. The model never misses a true landing — it's conservative in the safer direction for cost bidding.

Best Model & Conclusion

Best Model: Logistic Regression

Best params: C=1, L2 penalty, lbfgs solver

Why it wins: Ties SVM and KNN on raw accuracy (83.3%) but is far more interpretable — each feature's coefficient directly shows its effect on landing-success odds, which matters for explaining a cost bid to stakeholders. It's also the cheapest to retrain as new launches arrive.

Conclusion & Insights

- Landing success is highly learnable from launch parameters — 83% test accuracy from a simple, interpretable model.
- Orbit type and payload mass are the strongest visual predictors; heavier GTO missions land less reliably than light LEO/ISS missions.
- Success rate improved dramatically over time (2013 → 2020), reflecting SpaceX's iterative process improvements — not a static property of the hardware.
- Recommendation: a competing bidder can use this model as a first-pass cost-risk estimator, refining it further with more recent launches and continuous re-training.

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SpaceX Falcon 9 First-Stage Landing Prediction

Prepared by: Aaron Vargas
IBM Applied Data Science Capstone
August 19, 2026

Project Links

GitHub repository: <https://github.com/razerblade09/spacex-falcon9-capstone>

Presentation slides: the complete 19-slide presentation is included in this same PDF, immediately before this report.

Executive Summary

This project follows the full data science methodology — data collection, wrangling, exploratory analysis, interactive visualization, and predictive modeling — to solve a real business problem: predicting whether a SpaceX Falcon 9 first-stage booster will land successfully after launch.

Methods used: launch data was collected from the SpaceX REST API and by scraping Wikipedia's Falcon 9 launch history; the two sources were cross-checked and merged. The data was cleaned, missing values were imputed, and a binary landing-success label was engineered from raw outcome text. Exploratory analysis used both visualization (scatter plots, bar charts, a yearly trend line) and direct SQL querying. Two interactive tools — a Folium map and a Plotly Dash app — were built for stakeholder self-service exploration. Finally, four classifiers (Logistic Regression, SVM, Decision Tree, KNN) were trained and tuned with 10-fold GridSearchCV.

Key results: across 90 Falcon 9 launches (2010–Nov 2020), the overall first-stage landing success rate was 66.7%. Logistic Regression, SVM, and KNN all reached 83.3% test accuracy; Logistic Regression was selected as the best model for its interpretability. Success rate correlates strongly with orbit type (light LEO/ISS missions land far more reliably than heavy GTO missions) and improved steadily year over year as SpaceX matured its recovery process.

Introduction

Background: SpaceX advertises Falcon 9 launches at \$62 million, versus \$165 million or more from other providers, largely because SpaceX reuses the first stage instead of discarding it after every flight. If the first stage does not land successfully, the true cost of a launch changes substantially.

Problem statement: this project assumes the role of a data scientist for a startup competing with SpaceX. The goal is to predict, from historical and publicly available launch parameters, whether the first stage will land successfully — enabling an accurate cost estimate for a competing bid.

Objectives

- Collect and clean historical Falcon 9 launch data from multiple sources
- Identify which launch parameters correlate with landing success
- Visualize patterns across launch sites, orbits, and time
- Build and evaluate a machine-learning classifier for landing outcome

Data Collection – SpaceX API

The SpaceX v4 REST API (api.spacexdata.com) was queried for all past launches. Each launch record was filtered to single-core, single-payload Falcon 9 flights, then enriched by calling the /rockets, /launchpads,

/payloads, and /cores endpoints to resolve IDs into readable attributes (booster version, launch site name/coordinates, payload mass, orbit, landing outcome, grid fins, reused flag, legs, landing pad, booster block, and serial).

Result: 90 clean Falcon 9 launch records spanning 2010 through November 2020, one row per launch.

GitHub repository: https://github.com/razerblade09/spacex-falcon9-capstone/blob/main/notebooks/01_data_collection_api.py

Data Collection – Web Scraping

As a second, independent source, the Wikipedia page "List of Falcon 9 and Falcon Heavy launches" was scraped using requests and BeautifulSoup. HTML tables were located, column headers extracted programmatically, and each table row parsed into flight number, date/time, booster version, launch site, payload, payload mass, orbit, customer, launch outcome, and booster landing outcome.

This scraped dataset cross-validates the API data and was also loaded directly into a SQL database for the EDA-with-SQL phase below.

GitHub repository: https://github.com/razerblade09/spacex-falcon9-capstone/blob/main/notebooks/02_web_scraping.py

Data Wrangling Methodology

Missing values: PayloadMass was missing for 5.6% of records and was filled with the column mean. LandingPad was missing for 28.9% of records — this was left as-is, since a null landing pad legitimately indicates no ground/drone-ship landing occurred (e.g., an ocean landing or no landing attempt).

Label engineering: the raw Outcome field (free-text strings such as "True ASDS", "False Ocean", "None None") was converted into a single binary Class column: 1 if the booster landed successfully by any method (drone ship, ground pad, RTLS return, or ocean-controlled), 0 if it did not land or no attempt was made.

Result: of 90 launches, 60 (66.7%) resulted in a successful first-stage landing and 30 (33.3%) did not — this is the target variable for the predictive model.

GitHub repository: https://github.com/razerblade09/spacex-falcon9-capstone/blob/main/notebooks/03_data_wrangling.py

EDA with Data Visualization

Flight number, payload mass, orbit type, and launch year were plotted against landing outcome to surface which factors correlate with success.

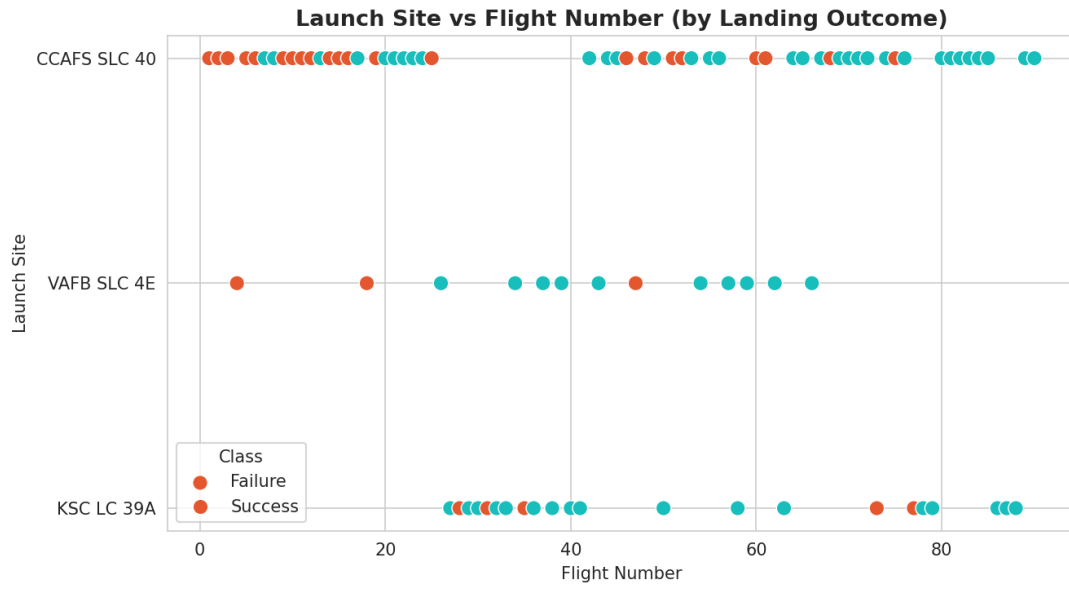


Figure 1. Flight number vs. launch site, colored by landing outcome.

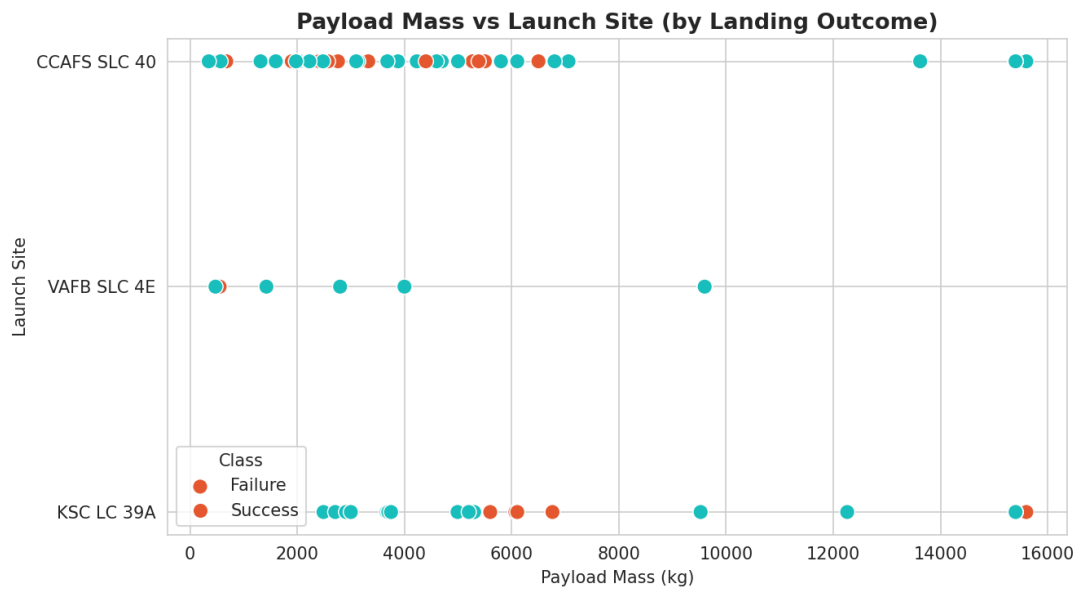


Figure 2. Payload mass vs. launch site, colored by landing outcome.

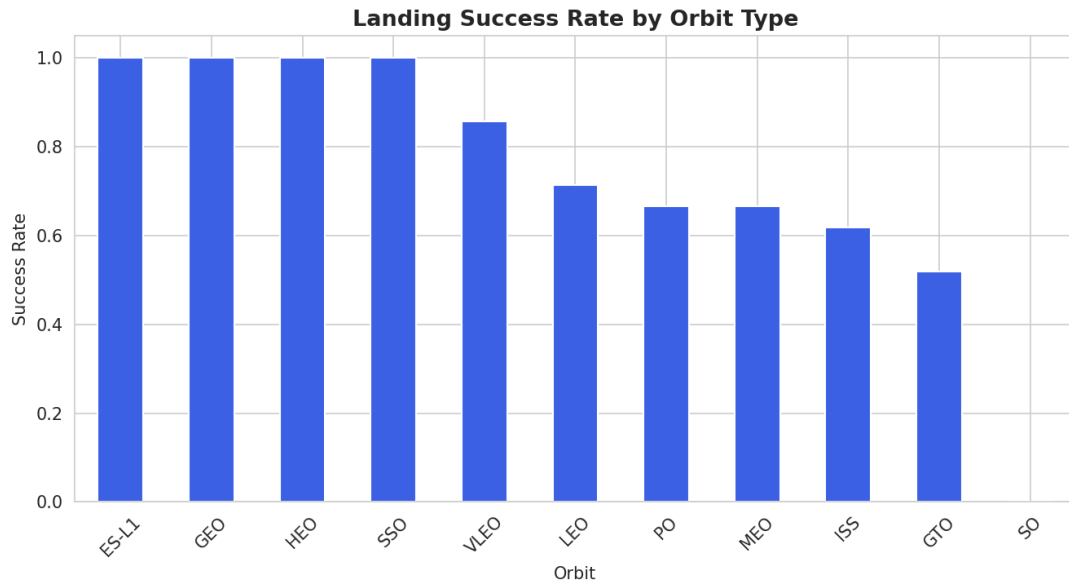


Figure 3. Landing success rate by orbit type.

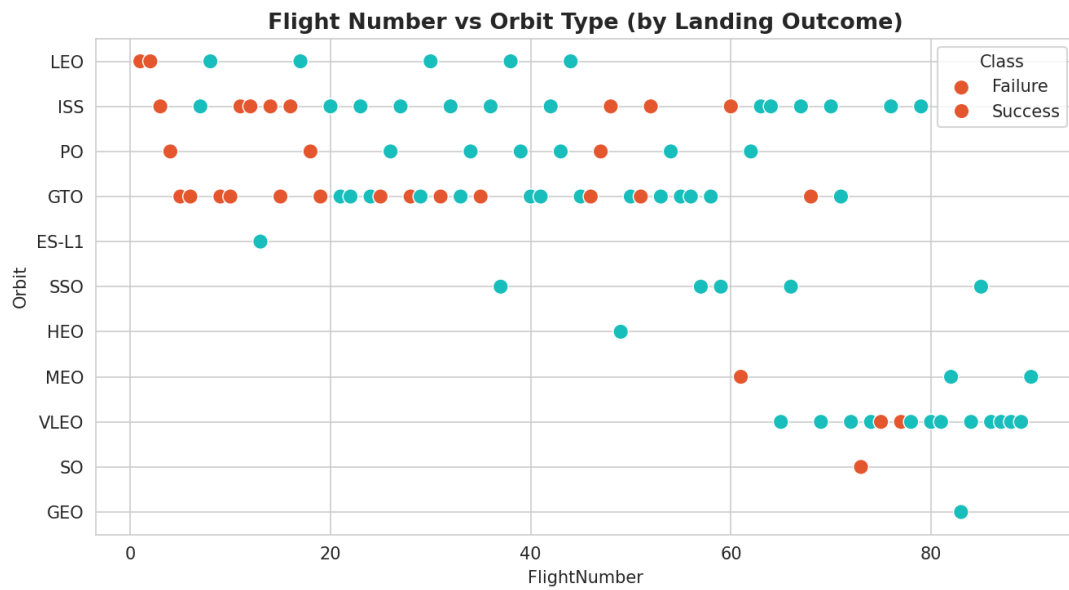


Figure 4. Flight number vs. orbit type, colored by landing outcome.

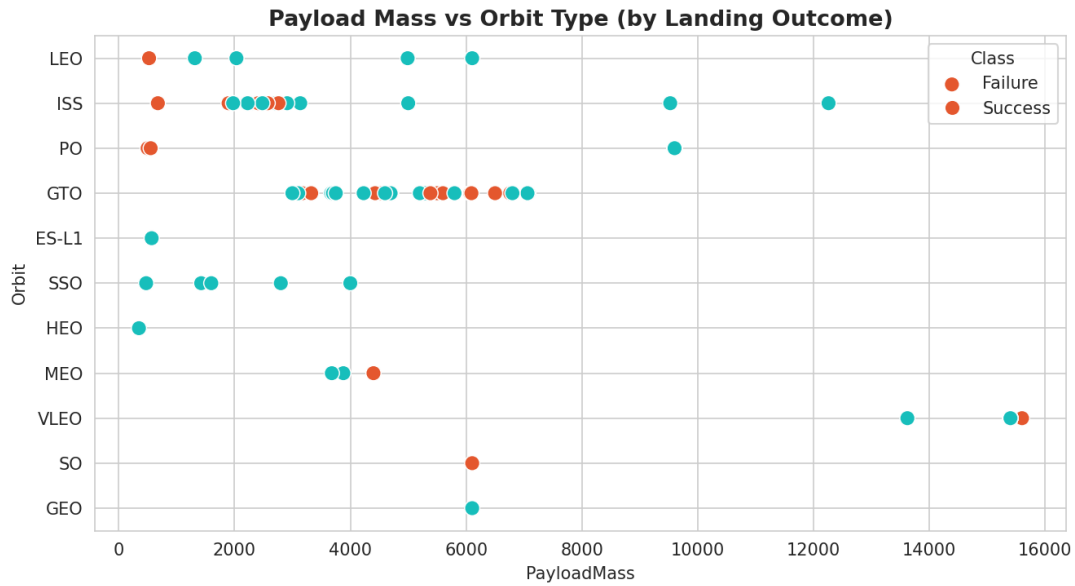


Figure 5. Payload mass vs. orbit type, colored by landing outcome.

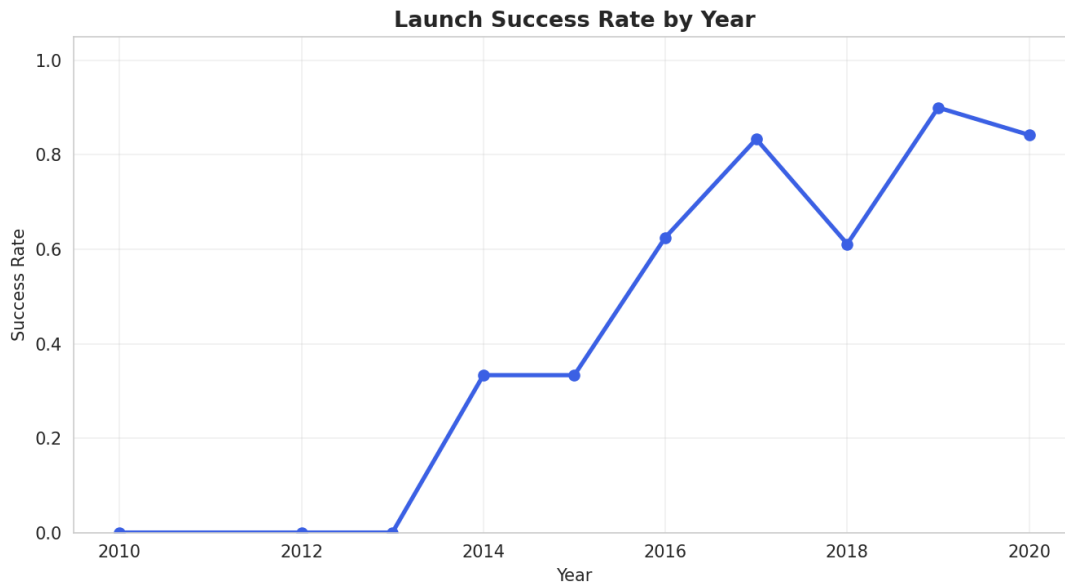


Figure 6. Landing success rate by year (2010–2020).

Summary: GTO and ISS-orbit missions (heavier, higher-energy trajectories) show lower success rates than LEO/SSO/GEO missions. Success rate climbed from near 0% in 2010–2013 to consistently near 100% by 2019–2020, reflecting SpaceX's iterative refinement of the recovery process rather than a static hardware property.

GitHub repository: https://github.com/razerblade09/spacex-falcon9-capstone/blob/main/notebooks/04_07_eda_sql_ml.py

EDA with SQL

The scraped launch table was loaded into a SQL database (SQLite for this report; IBM Db2 in the original lab environment) as SPACEXTABLE, then queried directly to answer specific business questions:

- Distinct launch sites used
- Total payload mass carried for NASA (CRS) missions
- Average payload mass for F9 v1.1 boosters
- Date of the first successful ground-pad landing
- Boosters that succeeded on a drone ship carrying 4,000–6,000 kg
- Count of each mission outcome (success/failure)
- Booster(s) carrying the maximum payload mass
- Failed drone-ship landings in 2015
- Ranking of landing outcomes between 2010 and 2017

Selected Results

Query	Result
Distinct launch sites	CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
Total payload for NASA (CRS)	45,596 kg
Avg. payload, F9 v1.1 boosters	2,534.7 kg
First successful ground-pad landing	1 May 2017
Max payload carried	15,600 kg (F9 B5)
Failed drone-ship landings in 2015	2 (F9 v1.1 B1012, B1015 — CCAFS LC-40)

Landing Outcome Ranking, 2010–2017

Landing Outcome	Count
Success	20
No attempt	10
Success (drone ship)	8
Success (ground pad)	6
Failure (drone ship)	4
Failure	3
Controlled (ocean)	3
Failure (parachute)	2

GitHub repository: https://github.com/razerblade09/spacex-falcon9-capstone/blob/main/notebooks/04_07_eda_sql_ml.py

Interactive Visual Analytics

Folium Map

An interactive Folium map plots all 4 launch sites and every individual launch (56 geocoded records), colored green for a successful landing and red for a failed/no-attempt landing.

SpaceX launch sites and landing outcomes — 56 launch records, 24 successful landings

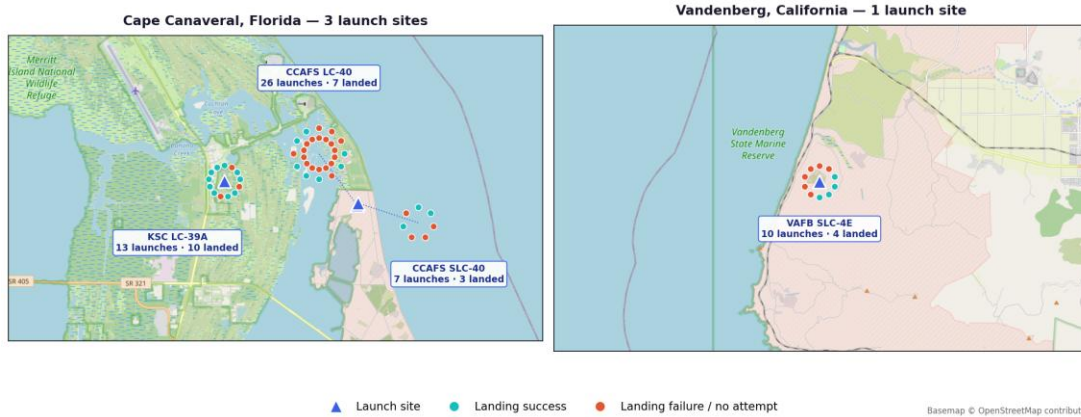


Figure 7. Static rendering of the interactive Folium map: all 4 launch sites with every launch record plotted and colored by landing outcome.

Proximity Analysis

Distances from KSC LC-39A to the nearest surrounding features were computed with the haversine formula. The coastline point was read off the map; the city, railway, and highway coordinates were taken from OpenStreetMap as the nearest place=city node and the nearest node of a way tagged railway=rail and highway=secondary respectively.

Feature	Nearest instance	Distance from KSC LC-39A
Coastline	Atlantic shoreline (east of pad)	7.43 km
City	Titusville, FL	16.33 km
Railway	NASA Railroad	0.69 km
Highway	Kennedy Parkway North	0.85 km

Proximity analysis — distances from KSC LC-39A

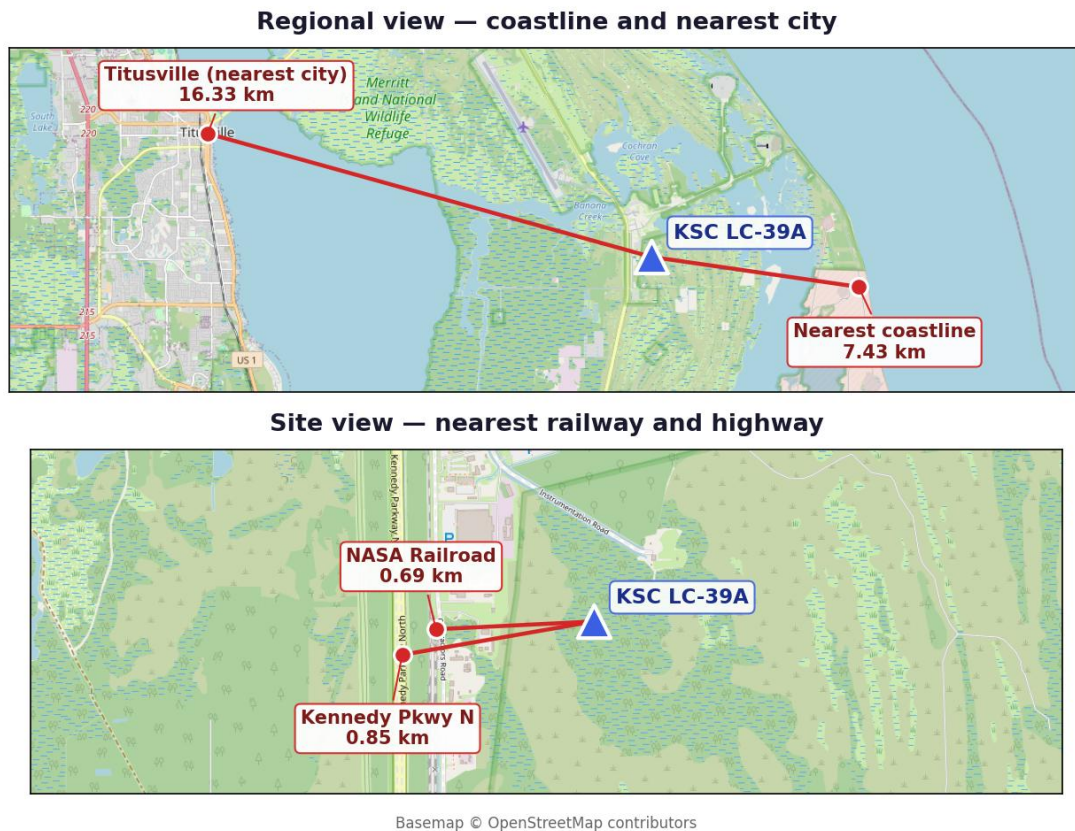


Figure 8. Proximity analysis — distance lines from KSC LC-39A to the nearest coastline, city, railway, and highway.

The pattern matches real launch-site siting constraints: the pad sits close to open water for downrange safety and within a kilometre of both rail and road links, since boosters and stages arrive overland, yet is kept well clear of the nearest population centre.

GitHub repository: https://github.com/razerblade09/spacex-falcon9-capstone/blob/main/notebooks/06_folium_map.py

Plotly Dash App

A Plotly Dash dashboard (dashboard/app.py) provides a dropdown to filter by launch site, a live-updating pie chart of success rate, and a payload-vs-outcome scatter plot filterable by a payload-mass range slider.

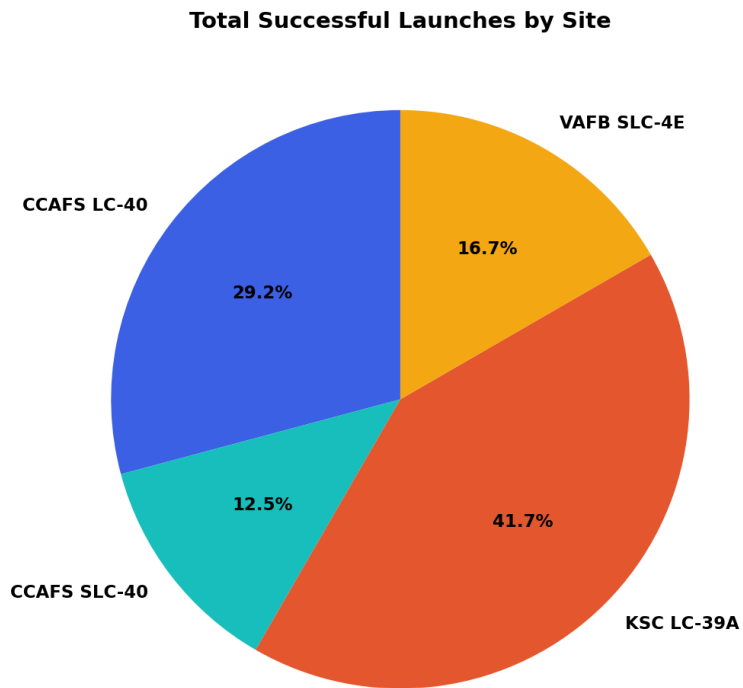


Figure 9. Share of total successful landings contributed by each launch site.

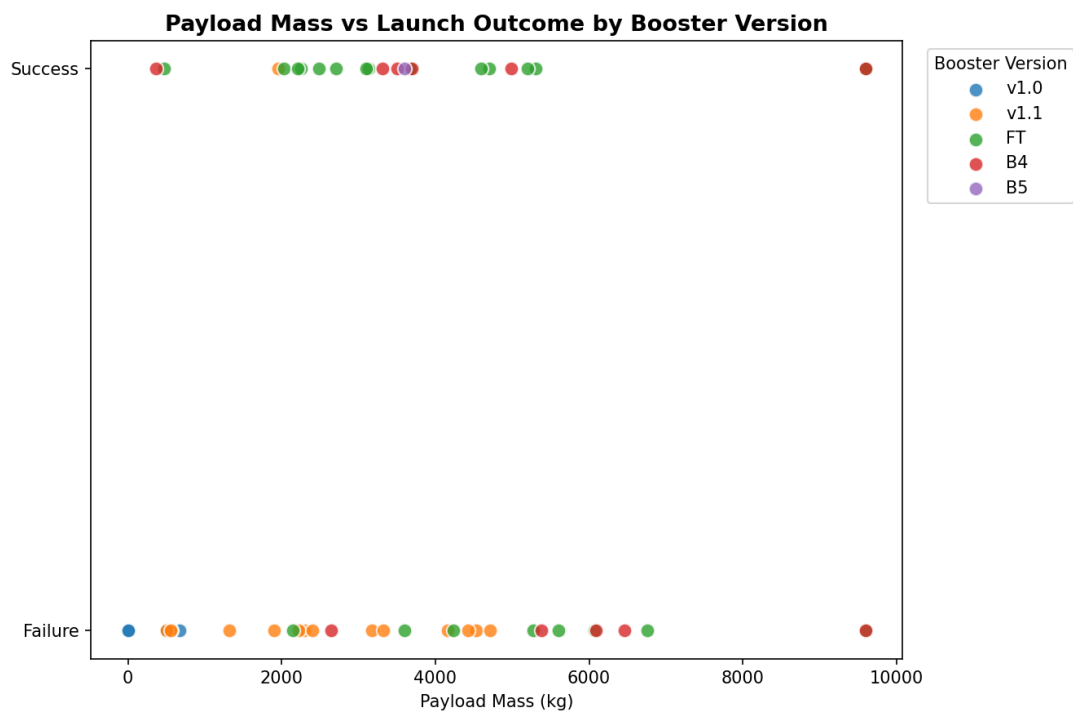


Figure 10. Payload mass vs. launch outcome, colored by booster version category.

Newer booster generations (FT, B4, B5) succeed reliably across a much wider payload range than early versions (v1.0, v1.1) — visual confirmation that booster maturity, not just payload mass alone, drives landing success.

GitHub repository: <https://github.com/razerblade09/spacex-falcon9-capstone/blob/main/dashboard/app.py>

Predictive Analysis

Methodology

Features: 80 one-hot-encoded columns (orbit, launch site, landing pad, booster serial) plus flight number, payload mass, flight count, grid fins, reused flag, legs, and booster block — standardized with StandardScaler. Data was split 80/20 into train (72 launches) and test (18 launches) sets.

Four classifiers were trained and tuned via 10-fold GridSearchCV: Logistic Regression (C, penalty, solver), Support Vector Machine (kernel, C, gamma), Decision Tree (criterion, max_depth, min_samples_split), and K-Nearest Neighbors (n_neighbors, distance metric).

Evaluation Results

Model	CV Score	Test Accuracy
Logistic Regression	0.821	0.833
SVM	0.834	0.833
KNN	0.834	0.833
Decision Tree	0.804	0.778

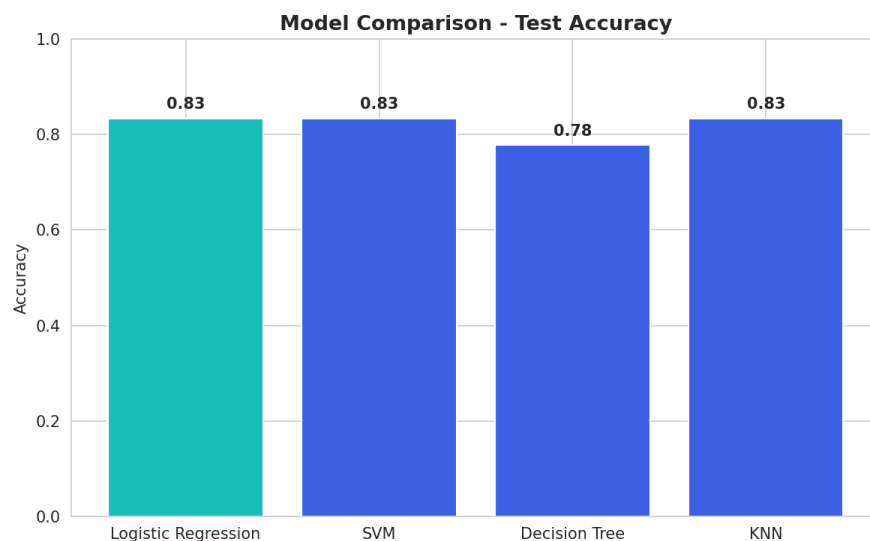


Figure 11. Test accuracy comparison across all four models.

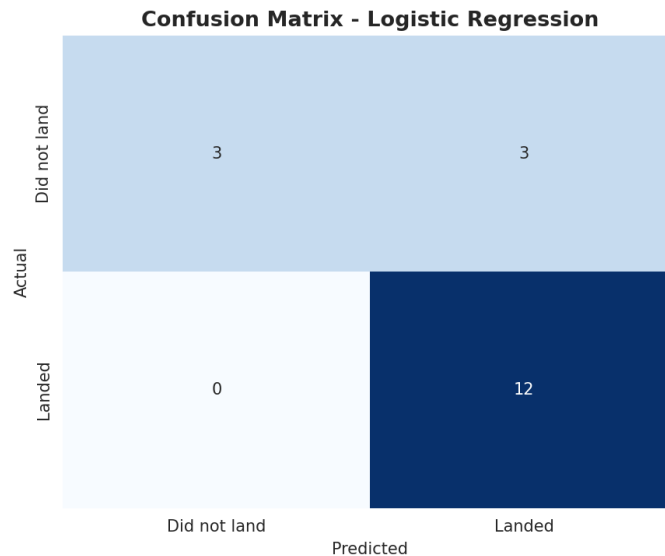


Figure 12. Confusion matrix for the best model (Logistic Regression).

Reading the confusion matrix (18 test launches): 12 successful landings correctly predicted, 3 unsuccessful landings correctly predicted, 3 unsuccessful landings misclassified as successful, and 0 successful landings missed. The model never misses a true landing, which is the more conservative error direction for cost estimation.

Best Model

Logistic Regression (C=1, L2 penalty, lbfgs solver) was selected as the best model. It ties SVM and KNN on raw test accuracy (83.3%) but is far more interpretable — each feature's coefficient directly shows its effect on landing-success odds, which matters when explaining a cost bid to non-technical stakeholders. It is also the cheapest of the four to retrain as new launch data becomes available.

Conclusion

- Landing success is highly learnable from launch parameters — 83.3% test accuracy from a simple, interpretable model.
- Orbit type and payload mass are the strongest predictors; heavier GTO missions land less reliably than light LEO/ISS missions.
- Success rate improved dramatically over time (2013 to 2020), reflecting SpaceX's iterative process improvements rather than a fixed hardware property.
- Recommendation: a competing launch provider can use this model as a first-pass cost-risk estimator for a Falcon-9-class bid, refining it further as more recent launch data becomes available and by incorporating weather and pad-specific factors in future work.